1 Planning (P)

Chlorine gas, when bubbled into hot dilute aqueous potassium hydroxide, produces a solution containing potassium chlorate(V) and potassium chloride.

(a) Write a balanced equation for the reaction between chlorine and potassium hydroxide.

$6KOH + 3Cl_2 \rightarrow KClO_3 + 5KCl + 3H_2O$ .... **[1]**

Evaporation of the resulting solution gives a white solid which is a mixture of potassium chlorate(V) and potassium chloride.

Pure potassium chloride can be separated from this mixture by a process known as **fractional crystallisation**. In this process, the substances are separated based on their difference in solubility.

The ***solubility*** of a solid is defined as:
'The mass of solid that will dissolve in and just saturate 100 g of solvent (water).'

(b) Most solids become more soluble as temperature increases.

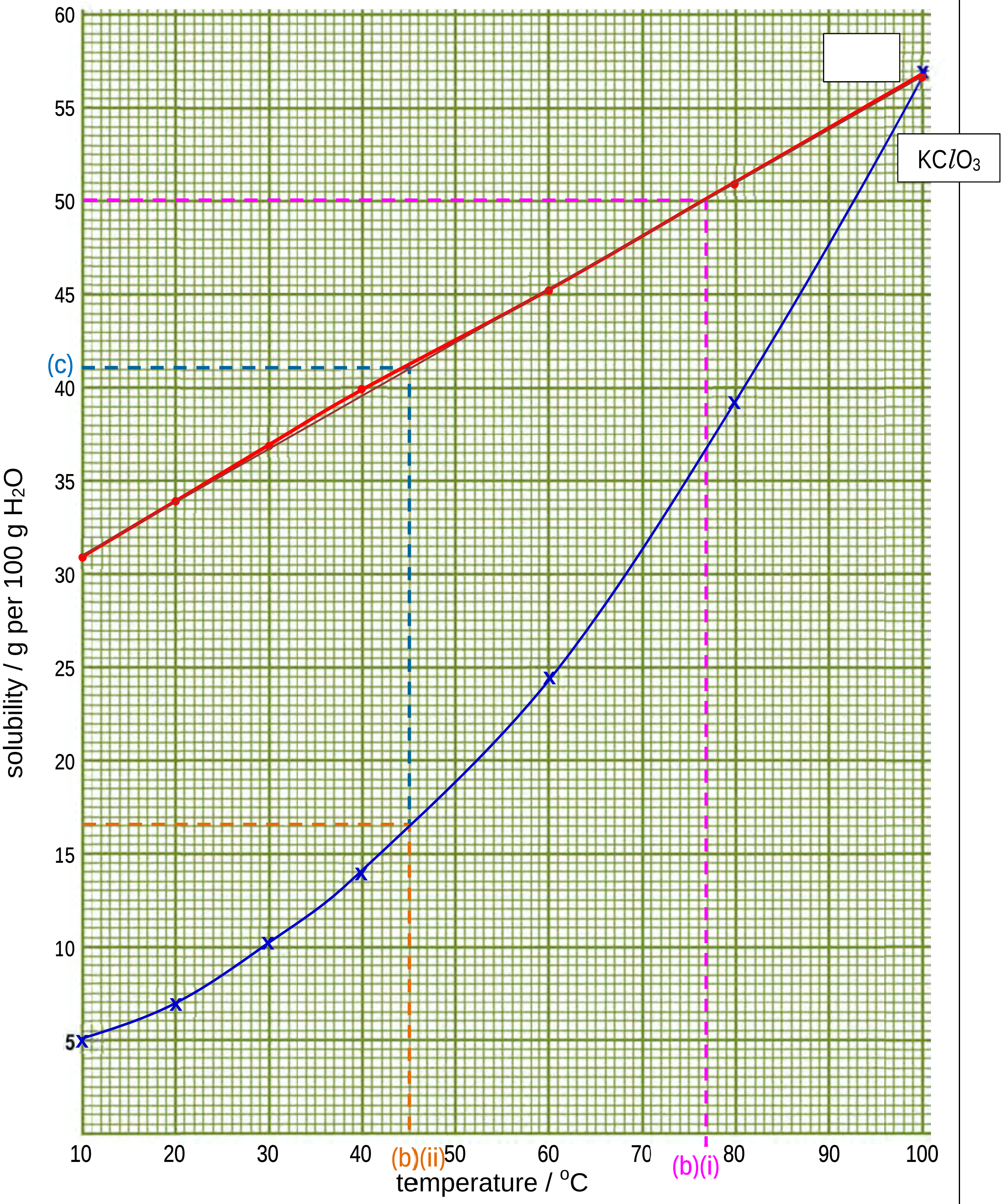
Table 1 below shows the solubility of potassium chlorate(V) and potassium chloride over a range of temperature.

temperature / ^{o}C	10	20	30	40	60	80	100
solubility of potassium chlorate(V), $KClO_3$, / g per 100 g of H_2O	5.0	7.0	10.5	14.0	24.5	38.5	57.0
solubility of potassium chloride, KCl, / g per 100 g of H_2O	31.0	34.0	37.0	40.0	45.5	51.1	56.7

Table 1

Plot the data in **Table 1** on the grid on the next page and draw the solubility curves for each substance.
Label the solubility curves clearly. **[3]**

For Examiners' Use



For Examiners' Use

(c) In the process of **fractional crystallisation**, the solid mixture is first dissolved in heated solvent to obtain a solution.
The resultant solution is then carefully cooled to a selected temperature to enable one solid to crystallise while ensuring the other solid remains dissolved in the solution.

Starting with a solid that contains **16.4 g** of potassium chlorate(V) and **50.0 g** of potassium chloride, you are required to use the solubility curves from **(a)** to plan a **fractional crystallisation** that will give you a sample of potassium chloride, free from potassium chlorate(V).
You may assume that all of the solid should be dissolved in 100 g of water at the start of the separation.

(i) From the graph in **(b)**, obtain the minimum temperature to which the solution must be heated in order to dissolve all of the 16.4 g of potassium chlorate(V) and 50.0 g of potassium chloride.

Temperature to which the solution must be heated : ..77.0 ± 1.0.... ^{o}C

(ii) Write a plan to obtain a <u>maximum</u> amount of potassium chloride, KCl, free from potassium chlorate(V), $KClO_3$, from the mixture that contains **16.4 g** of potassium chlorate(V) and **50.0 g** of potassium chloride .

You may assume that you are provided with the apparatus normally found in a school or college laboratory.

Your plan should include:
- the essential details of the general fractional crystallisation procedure, including the capacity of apparatus used (where relevant)
- the temperature to which the solution is to be cooled and the justifications for your choice (reference to the graph in **(b)** is required)
- a precautionary step you would take to improve the purity of the potassium chloride sample crystals that are obtained

1) Place the solid mixture into a 250 cm³ conical flask, add 100 g of water and heat to about 80°C *(temp from part (i)* in a water bath.
2) Cool the solution to 45°C in a water bath.
3) Filter the mixture to obtain the KCl crystals in the residue.

To maintain the solution at 45°C during the filtration process and minimise crystallization of $KClO_3$, <u>the filter paper, funnel and conical flask should be rinsed with water (or alcohol) of the same temperature</u> (or slightly higher).

Justification for choice of 45°C:
From the graph in (a), in order to ensure that all 16.4 g of $KClO_3$ remains dissolved, the temperature should not be below 45.0°C. [Working line shown on graph.]

- general procedure includes: dissolving/cooling/filtration
- appropriate choice of apparatus (conical flask with capacity stated (accept 150 cm³ capacity; choice of water bath to maintain temperature rather than direct heating)
- due regard for reliability: maintains temperature during filtration by rinsing filtration apparatus with warm solvent

Justification and choice of temperature :
- construction lines shown on graph for temperature value / value read correctly (±0.5) and used correctly in procedure

[4]

over

For Examiners' Use

....

....

....

....

[5]

(d) With reference to the graphs in **(b)**, calculate the mass of potassium chloride, KCl, you would expect to recover.

From the graph, at 45°C, mass of KCl still dissolved = 41.0 g [1]
[Construction lines shown on graph.]

Therefore, mass of KCl recovered = 50.0 – 41.0 = 9.0 g [1]

(e) How could you check the purity of the sample of potassium chloride that has been obtained?

If the sample melts at a fixed melting point (of KC*l*), it is pure. [1]

....

[1]

2 The following table lists the standard enthalpy changes of combustion, $\Delta H_c^{\ominus}$, of some monohydric alcohols.

alcohol	**$\Delta H_c^{\ominus}$ / kJ mol^{-1}**
methanol	−715
ethanol	−1367
propan-1-ol	
butan-1-ol	−2671

For Examiners' Use

(a) Write an equation to represent the standard enthalpy change of combustion of propan-1-ol.

$CH_3CH_2CH_2OH(l) + 9/2\ O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$ **[1]**

(b) **(i)** The difference in the standard enthalpy change of combustion of methanol and ethanol is −652 kJ mol^{-1}. By considering the structures of the two alcohols, suggest the significance of this difference.

It is the enthalpy change of combustion of a $-CH_2-$ group.

(ii) Hence suggest a value of $\Delta H_c^\ominus$ for propan-1-ol.

$\Delta H_c^\ominus$ for propan-1-ol = –2019 kJ mol^{-1}

(iii) Given $\Delta H_f^\ominus(H_2O(l)) = -286$ kJ mol^{-1} and $\Delta H_f^\ominus(CO_2(g)) = -394$ kJ mol^{-1} calculate the standard enthalpy change of formation of propan-1-ol, $\Delta H_f^\ominus(CH_3CH_2CH_2OH(l))$.

$$\Delta H_c^\ominus(CH_3CH_2CH_2OH(l)) = 3\Delta H_f^\ominus(CO_2(g)) + 4\Delta H_f^\ominus(H_2O(l)) - \Delta H_f^\ominus(CH_3CH_2CH_2OH(l))$$

$$-2019 = [3(-394) + 4(-286)] - \Delta H_f^\ominus(CH_3CH_2CH_2OH(l))$$

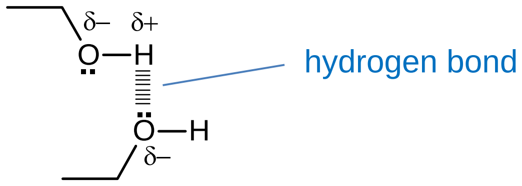
$$\Delta H_f^\ominus(CH_3CH_2CH_2OH(l)) = -307 \text{ kJ mol}^{-1}$$

[4]

[Turn over

For Examiners' Use

(c) Draw a labelled diagram to show the interactions between ethanol molecules.



[2]

[Total: 7 marks]

3 (a) A student electrolyses a solution of brine (concentrated aqueous sodium chloride) using platinum electrodes. She ensures that the solution is continually stirred at a temperature of not more than 40^{o}C.

It was found that as electrolysis proceeds, sodium chlorate(I), Na*Cl*O, is formed in the electrolyte.

(i) Suggest equations for the reactions occuring at the anode and cathode.

anode: $2Cl^- \rightarrow Cl_2 + 2e^-$

cathode: $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$

(ii) Write a suitable equation which shows how the Na*Cl*O could have been formed.

$Cl_2 + 2NaOH \rightarrow NaClO + NaCl + H_2O$

[3]

For Examiners' Use

(b) Sodium chlorate(I), $NaClO$, which is found in household cleaning products, is usually manufactured from chlorine gas.

When gaseous phosphorous (V) chloride is heated to 473 K in a closed container, the following equilibrium involving chlorine gas is established:

$$PCl_5\,(g) \rightleftharpoons PCl_3\,(g) + Cl_2\,(g) \qquad \Delta H = > 0$$

By using Le Chatelier's principle, predict and explain the effect of an increase in temperature on the above equilibrium.

- By LCP, when temperature increases, the system favours an endothermic reaction in order to reduce the temperature.
- Since the forward reaction is endothermic, the equilibrium shifts right.

[2]

(c) When sodium thiosulfate is reacted separately with chlorine and iodine, different products are formed.

$$4Cl_2 + Na_2S_2O_3 + 5H_2O \rightarrow 2NaHSO_4 + 8HCl$$

$$I_2 + 2Na_2S_2O_3 \rightarrow Na_2S_4O_6 + 2NaI$$

(i) With reference to the change in oxidation state of sulfur in the reactions, deduce the relative oxidising strengths of chlorine and iodine.

- In the reaction with Cl_2, the oxidation state of S has increased from +2 in $S_2O_3^{2-}$ to +6 in HSO_4^- .
- In the reaction with I_2, the oxidation state of S has increased from +2 in $S_2O_3^{2-}$ to +2.5 in $S_4O_6^{2-}$.
- Since Cl_2 has caused a greater increase in oxidation state of the sulphur, Cl_2 is a stronger oxidising agent than iodine.

[Turn over

For Examiners' Use

(ii) Use the following data, and data from the *Data Booklet*, to suggest the sulfur containing species that will be formed when bromine reacts with sodium thiosulfate.

$S_4O_6^{2-} + 2e^- \rightleftharpoons 2\ S_2O_3^{2-}$ $E^\ominus = +0.09$ V

$4SO_2 + 4H^+ + 6e^- \rightleftharpoons S_4O_6^{2-} + 2H_2O$ $E^\ominus = +0.51$ V

$SO_4^{2-} + 4H^+ + 2e^- \rightleftharpoons SO_2 + 2H_2O$ $E^\ominus = +0.17$ V

For Br_2 to oxidise $S_2O_3^{2-}$ to $S_4O_6^{2-}$, $E^\ominus_{cell} = +1.07 – 0.09 = +0.98$ V
For Br_2 to further oxidise $S_4O_6^{2-}$ to SO_2, $E^\ominus_{cell} = +1.07 + (–0.51) = +0.56$ V.
Once SO_2 has been formed, for Br_2 to further oxidise SO_2 to SO_4^{2-}, $E^\ominus_{cell} = +1.07 –0.17 = 0.90$ V
Since **all the** $E^\ominus_{cell} > 0$, the reactions are expected to be feasible.

Therefore the final product is: SO_4^{2-} **or** HSO_4^-

……………………………………………………………………………………

……………………………………………………………………………………

……………………………………………………………………………………

[4]

[Total: 9]

For Examiners' Use

4 Chromium is a typical transition element. Many of its compounds are intensely coloured.

(a) Write the electronic configuration for the chromium atom and chromium (III) ion.

Cr : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$

Cr^{3+} : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$ **[2]**

(b) A reaction scheme involving chromium compounds is shown below:

$$\underset{\mathbf{A}}{\boxed{Cr_2O_7^{2-}}} \xrightarrow[\textbf{Step 1}]{SO_2} \underset{\mathbf{B}}{\boxed{[Cr(H_2O)_6]^{3+}}} \xrightarrow[\textbf{Step 2}]{H_2\ddot{N}CH_2CH_2\ddot{N}H_2} \boxed{\mathbf{C}}$$

The reaction in **Step 2,** which occurs spontaneously, involves 3 moles of the ligand reacting with 1 mole of **B** to form **C**.

(i) Explain why chromium compounds are coloured.

- The five d orbitals of an isolated transition metal gaseous atom have the same energy. The presence of ligands in transition metal compounds causes the splitting of energy levels of the *d* orbitals in the transition metal ion into two or more groups, with an energy gap (ΔE) between them.
- Electrons in a lower energy *d*-orbital can jump to a higher energy level by absorbing light energy. This process is known as d→d electronic transition.
- The observed colour is the complement of the colour absorbed.

(ii) What type of reaction occurs in **Step 1**?

reduction

(iii) Name the type of bond formed between the metal ion and the ligand in **Step 2**.

dative (or co-ordinate) bond

(iv) Write an equation for the reaction in **Step 2**.

$[Cr(H_2O)_6]^{3+} + 3\ H_2NCH_2CH_2NH_2 \rightleftharpoons [Cr(H_2NCH_2CH_2NH_2)_3]^{3+} + 6H_2O$

For Examiners' Use

(v) By considering the spontaneity of the reaction, together with the equation in **(b)(iv)**, deduce the signs of ΔS, ΔH and ΔG for the reaction in **Step 2**.

- There is an increase in entropy when the above equilibrium shifts right as six molecules of H_2O are released from the complex, therefore ΔS will be positive.
- ΔH will be negative because the bonds that are being formed between the ligand and the Cr^{3+}ion are stronger than those that are being broken. (or $H_2NCH_2CH_2NH_2$ forms stronger bonds with Cr^{3+} than those formed by water.)
- ΔG will be negative because the reaction is described as being spontaneous.

[9]

(c) Silver chromate(VI) is a chromium containing compound which has applications in the field of photography.

The solubility product of Ag_2CrO_4 at 25°C is 1.12×10^{-12} $mol^3\ dm^{-9}$.

(i) Write an expression for the solubility product, K_{sp}, of Ag_2CrO_4.

K_{sp} of Ag_2CrO_4 = $[Ag^+]^2[CrO_4^{2-}]$

(ii) Calculate the solubility, in $mol\ dm^{-3}$, of Ag_2CrO_4, at 25 °C.

Let the solubility be x $mol\ dm^{-3}$

$K_{sp} = (2x)^2(x)$
$4x^3 = 1.12 \times 10^{-12}$
$x = 6.54 \times 10^{-5}$

Solubility = 6.54×10^{-5} $mol\ dm^{-3}$

(iii) Concentrated aqueous silver nitrate is added dropwise to 0.010 $mol\ dm^{-3}$ potassium chromate(VI).

What is the concentration, in $mol\ dm^{-3}$ of silver ions when the first trace of precipitate appears?

Let the required $[Ag^+]$ be y $mol\ dm^{-3}$

$K_{sp} = (y)^2(0.010)$
$y = 1.06 \times 10^{-5}$

$[Ag^+] = 1.06 \times 10^{-5}$ $mol\ dm^{-3}$

[3]

[Total: 14]

For Examiners' Use

5 Compound **X** is a disubstituted aromatic compound with the molecular formula $C_8H_9O_2N$ and contains 2 functional groups.

Further investigations about compound **X** were conducted and the data about the reactions of **X** is shown in the table.

reaction	reagent	result
1	NaOH at room temperature	colourless solution formed
2	HC*l* at room temperature	no reaction
3	NaOH under reflux followed by careful acidification	two products were formed; one product had M_r of 152 and while the other was a gas that turned red litmus blue
4	Na	colourless gas evolved; white solid formed which is soluble in polar solvents
5	Br_2 (aq) in excess	Br_2 is decolourised; white solid formed with M_r = 308.8

Based on the information in the table and in this question, answer the following.

(a) **(i)** Name the functional group that both reaction **1** and **5** show to be present in **X**.

phenol

(ii) Name the functional group present in **X** based on the results obtained in **both** reactions **2** and **3**. Explain your answer using **either** reaction **2 or 3**.

Primary amide (or amide).
X was unreactive towards a strong acid (HCl). Hence, the functional group is amide and not amine as amines are basic and will react with HCl (or amides are neutral).
OR
X contains a primary amide and when subjected to base hydrolysis, it formed ammonia which turned red litmus blue.

[3]

(b) Deduce the molecular formula of the white solid formed in reaction **5**.

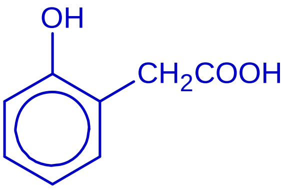
$C_8H_7O_2NBr_2$
(to support with comment about difference in M_r indicating incorporation of 2 Br atoms)

[1]

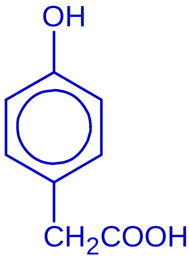
[Turn over

For Examiners' Use

(c) Draw a possible structural formula for the organic product in reaction **3**.

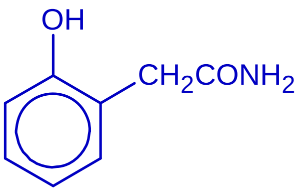


or

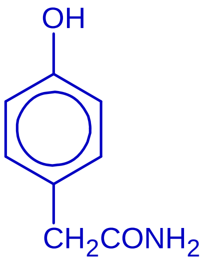


[1]

(d) **(i)** Use the given information, together with your answers in parts **(a)** to **(c)** , draw the structural formula of **X**.



or



(ii) Explain clearly why you have placed **each** of the two functional groups in their particular positions.

The –OH functional group is 2,4-directing in the reaction wtih bromine.

Since only 2 Br atoms are introduced in reaction 5, the 4th position (or 2nd position) with respect to –OH functional group has to be occupied by the other (amide) functional group in X.

[3]

[Total: 8]

For Examiners' Use

6 Hydrocarbons are economically important chemicals due to their role as the world's main source of fuel, and as starting materials for the synthesis of many useful organic compounds.

(a) The table below gives the boiling point of three organic compounds of similar molecular mass.

Organic Compound	Structural Formula	M_r	Boiling point/ °C
hexane	$CH_3CH_2CH_2CH_2CH_2CH_3$	86.0	68
2-methylpentane	$CH_3CH(CH_3)CH_2CH_2CH_3$	86.0	60
pentan-2-ol	$CH_3CH(OH)CH_2CH_2CH_3$	88.0	120

Explain the difference in boiling point between

(i) hexane and 2-methylpentane

- Hexane molecules are more elongated, and have greater surface area for interaction between molecules than 2-methylpentane molecules.
- Stronger induced dipole-dipole attraction between hexane molecules than 2-methylpentane, resulting in more energy required to separate hexane molecules (or overcome the intermolecular forces).

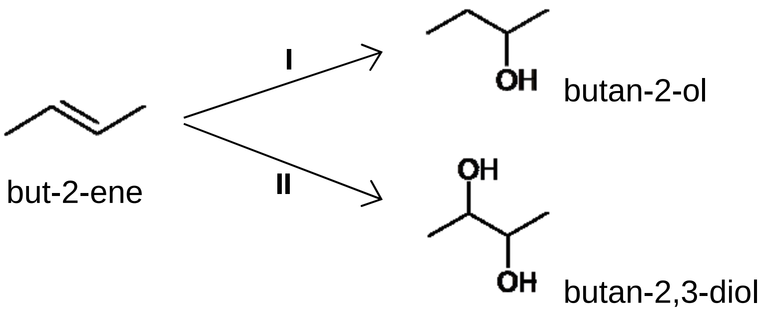
(ii) 2-methylpentane and pentan-2-ol

- The main intermolecular interaction between pentan-2-ol molecules is hydrogen bonding, while that between 2-methylpentane molecules is induced dipole – induced dipole attraction.
- Since hydrogen bonding is stronger than induced dipole – induced dipole attraction, more energy is required to break the stronger hydrogen bonding between pentan-2-ol molecules, resulting in a higher boiling point.

[4]

(b) Alkenes of low molecular mass can be produced via the catalytic cracking of long chain hydrocarbons.

In the presence of suitable reagents, but-2-ene can react to produce alcohols such as butan-2-ol and butan-2,3-diol.



[Turn over

For Examiners' Use

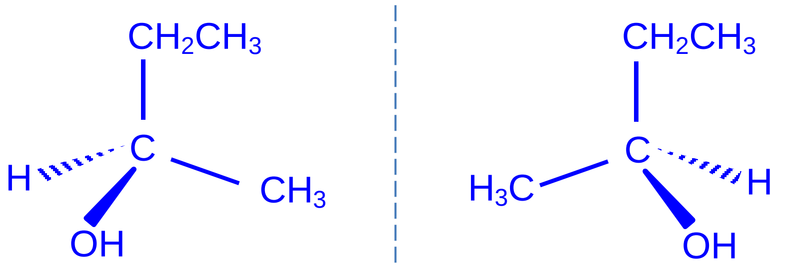
(i) Suggest suitable reagents and conditions for the conversion of but-2-ene to

butan-2-ol .... cold concentrated H_2SO_4,followed by adding water and warm OR steam with H_3PO_4 catalyst at 300^oC, 70 atm

butan-2,3-diol .... cold alkaline $KMnO_4$..

Butan-2-ol exhibits optical isomerism. However, the product obtained from step **I** does not rotate plane polarised light.

(ii) Draw structural formulae to illustrate the optical isomers of butan-2-ol.



[1]

(iii) Suggest why the butan-2-ol obtained in step **I** does not rotate plane polarised light.

... A racemic mixture containing an equal proportion of the two optical isomers of butan-2-ol is obtained, and hence does not rotate plane polarised light. ...

The effect of plane polarised light on butan-2,3-diol was investigated.

Three different types of butan-2,3-diol molecules were identified.

- Molecule **A** rotated plane polarised light to the left.
- Molecule **B** rotated plane polarised light to the right.
- Molecule **C** had no effect on plane polarised light.

(iv) Suggest an explanation for the above observations made for molecules **A** and **C**.

Molecule	Explanation
A	The arrangement of groups at both chiral carbons is such that plane polarised light is rotated to the left. Hence, the overall effect is that plane polarised light is rotated to the left.
C	The arrangement of groups at the chiral carbons is such that plane polarised light is rotated to the right at one chiral carbon, but rotated to the left at the other chiral carbon. The rotation of light to the right exactly cancels that to the left, resulting in no rotation overall. (Or extent of rotation is equal for both carbons)

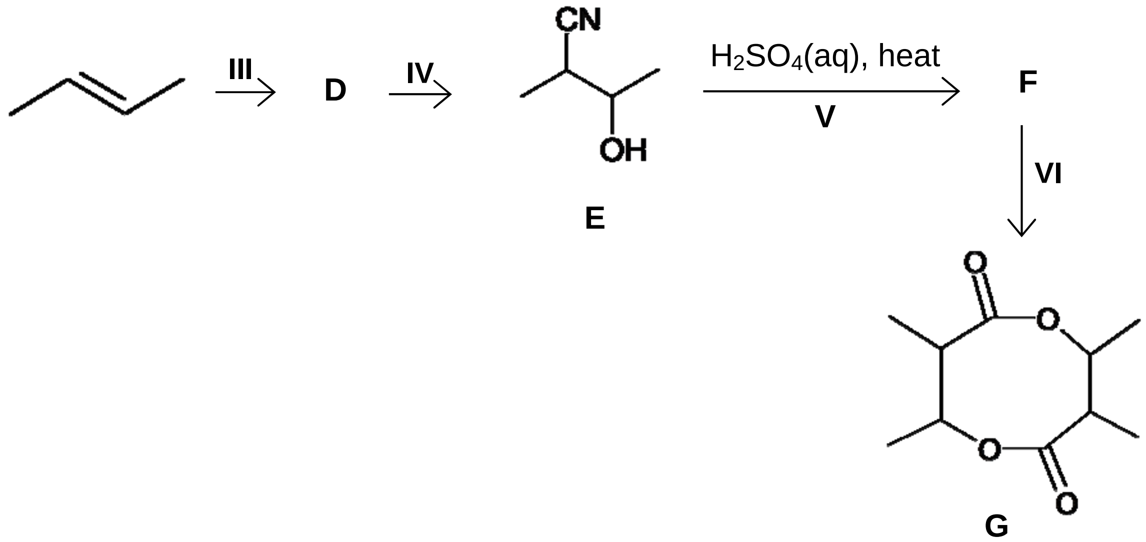
For Examiners' Use

(v) Suggest the ratio in which molecules of **A**, **B** and **C** are formed.

Ratio of **A**: **B**: **C** 1 : 1 : 2

[7]

(c) The following reaction scheme shows how but-2-ene can be used to synthesize a cyclic diester, **G**.



(i) Suggest suitable reagents and conditions for steps **III**, **IV** and **VI**.

Step **III** aqueous bromine, (r.t.p)

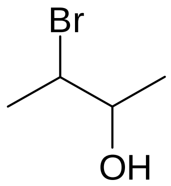
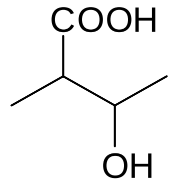
Step **IV** alcoholic KCN, reflux

Step **VI** concentrated H_2SO_4, reflux

(ii) State the type of reaction occurring in step **V**.

(acid) hydrolysis

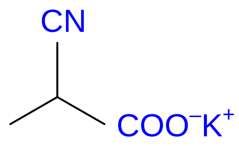
(iii) Suggest the structures of compounds **D** and **F**.

D	**F**
	

[Turn over

For Examiners' Use

(iv) Give the structural formula of the organic products formed when **E** is warmed with a solution of iodine and potassium hydroxide.



CHI_3

(v) But-1-ene is an isomer of but-2-ene.

Describe a chemical test that would enable you to distinguish between but-1-ene and but-2-ene.
Your answer should include the reagents and conditions for the test as well as relevant observations. Write balanced equations for any reaction that occurs.

Test: Add acidified $KMnO_4$ and heat.

Observations:

- But-1-ene: Purple $KMnO_4$ decolourised, and effervescence observed.
- But-2-ene: Purple $KMnO_4$ decolourised. (no effervescence)

Equations:

- But-1-ene: $CH_3CH_2CH{=}CH_2 + 5\ [O] \longrightarrow CH_3CH_2COOH + CO_2 + H_2O$
- But-2-ene: $CH_3CH{=}CHCH_3 + 4\ [O] \longrightarrow 2\ CH_3COOH$

[11]

[Total: 22]

----- End of Paper -----